C1.2 Why are there temperature changes in chemical reactions?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*	*Linked learning opportunities*
When a fuel is burned in oxygen the surroundings are warmed; this is an example of an exothermic reaction. There are also chemical reactions that cool their surroundings; these are endothermic reactions. Energy has to be supplied before a fuel burns. For all reactions, there is a certain minimum energy needed to break bonds so that the reaction can begin. This is the activation energy. The activation energy, and the amount of energy associated with the reactants and products, can be represented using a reaction profile. Atoms are rearranged in chemical reactions. This means that bonds between the atoms must be broken and then reformed. Breaking bonds requires energy (the activation energy) whilst making bonds gives out energy. **Energy changes in a reaction can be calculated if we know the bond energies involved in the reaction.**	1. distinguish between endothermic and exothermic reactions on the basis of the temperature change of the surroundings 2. draw and label a reaction profile for an exothermic and an endothermic reaction, identifying activation energy 3. explain activation energy as the energy needed for a reaction to occur 4. interpret charts and graphs when dealing with reaction profiles 5. **calculate energy changes in a chemical reaction by considering bond breaking and bond making energies** M1a, M1c, M1d 6. carry out arithmetic computations when calculating energy changes M1a, M1c, M1d 7. describe how you would investigate a chemical reaction to determine whether it is endothermic or exothermic *(separate science only)*	**Practical work:** • investigate different chemical reactions to find out if they are exothermic or endothermic
Using hydrogen fuel cells as an alternative to fossil fuels for transport is one way to decrease the emission of pollutants in cities (IaS4). The reaction in the fuel cell is equivalent to the combustion of hydrogen and gives the same product (water) but the energy drives an electric motor rather than an internal combustion engine. However, hydrogen is usually produced by electrolysis, which may use electricity generated from fossil fuels so pollutants may be produced elsewhere. There are difficulties in storing gaseous fuel for fuel cells which limits their practical value for use in cars.	8. recall that a chemical cell produces a potential difference until the reactants are used up *(separate science only)* 9. evaluate the advantages and disadvantages of hydrogen/ oxygen and other fuel cells for given uses *(separate science only)*	**Ideas about Science:** • fuel cells as a positive application of science to mitigate the effects of emissions (IaS4)